

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech. (CL) Examination****March – April, 2018****CL313/313.01 Hydrology & Ground Water Hydraulics****Date: 05.04.2018, Thursday****Time: 01:30 p.m. To 4.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Select the correct answer for the following and rewrite the complete correct statement: [7]

- (a) The percentage of earth covered by ocean is about
(a) 31% (b) 51% (c) 71% (d) 97%
- (b) The water year in India starts from the first day of
(a) January (b) April (c) June (d) July
- (c) A plot between rainfall intensity v/s time is called as
(a) Hydrograph (b) mass curve (c) hyetograph (d) isohyte
- (d) Water flows over the land and is first to reach the streams and rivers is known as
(a) base flow (b) sub-surface runoff (c) precipitation (d) surface runoff
- (e) A unit hydrograph has
(a) one unit of peak discharge (b) one unit of direct runoff
(c) one unit of time base of direct runoff (d) one unit of rainfall duration
- (f) Evapotranspiration is confined
(a) to a day light hours (b) night time only
(c) land surface only (d) none of these
- (g) For an annual flood series arranged in descending order of magnitude, the return period for a magnitude listed at position m in a total of N data entries is
(a) m/N (b) $m/(N+1)$ (c) $(N+1)/m$ (d) $N/(m+1)$

Q - 2 (i) Define: Hydrology. Draw a neat sketch of hydrologic cycle. [3]

- (a) (ii) A catchment area has six rain gauge stations. In a year the annual rainfall recorded by the gauges are as follows: [4]**

Station	A	B	C	D	E	F
Rainfall (cm)	82.6	102.9	180.3	110.3	98.8	136.7

For 10% error in estimation of mean rainfall, calculate the minimum number of additional rain gauge stations to be established in the catchment area.

- (b) (i) Explain 'evaporation'. Discuss the factors affecting evaporation. [4]**
(ii) Define Precipitation. Enlist forms of precipitation. [3]

OR

- (b) (i) Explain with a neat sketch : Measurement of infiltration with a single and double ring type infiltrometer [4]**
(ii) Normal annual rainfall at stations A, B, C and D are 60.97, 65.28, 78.12 and 90.14 cm respectively. In a year 1985 the D station was inoperative and annual precipitation [3]

recorded at A, B, C is 81.12, 68.13 and 75.50. Estimate the annual rainfall at station D in that year.

- Q – 3** (i) What is salt water intrusion? How can it be controlled? [4]
 (a) (ii) Explain the terms: Infiltration, Infiltration capacity, Base flow. [3]
 (b) (i) What is Runoff? Discuss the factors affecting the runoff coefficient. [4]
 (ii) Enlist applications of hydrology in civil engineering? [3]

OR

- (b) (i) Explain: Unit Hydrograph (UH). What are the assumptions in UH theory? [3]
 (ii) Given the ordinates of a 4-hr unit hydrograph as below. Derive the ordinates of a 12-hr unit hydrograph for the same catchment by S-curve method. [4]

Time (h)	0	4	8	12	16	20	24	28	32	36	40	44
Ordinate of 4-h UH (m ³ /s)	0	20	80	130	150	130	90	52	27	15	5	0

SECTION – II

- Q.4** Explain in brief [7]
 (1) Specific yield (2) Storage coefficient (3) Specific capacity (4) Confined Aquifer
 (5) Water table (6) Safe yield of an aquifer (7) Piezometric surface

Q.5

- (a) The average thickness of the confined aquifer is 20 m and extends over an area of 720 km². The piezometric surface fluctuates annually from 22 to 15 m above the top of aquifer. What ground water can be expected annually, if a storage coefficient is 0.00072? Assuming, an average well yield of 20 m³/hr and about 100 days of pumping in a year, how many wells can be drilled in the area? [7]
 Comment on your answer considering the pumping rate is increased in the area.

OR

- (a) A 30 cm well fully penetrates a 30 m deep confined aquifer. After a long period of pumping at the rate 1200 lpm the drawdown in wells 20 and 45 m from pumping well is observed as 2.2 and 1.8 m respectively. [7]
 (b) What is salt water intrusion? Enlist the measures that can be taken to control salt water intrusion? [5]
 State Darcy's Law governing groundwater movement. [2]

Q.6

- (a) Determine the transmissibility of the aquifer. What is the drawdown in the pumped well? [5]
 Enlist the different methods of artificial recharge of groundwater. [2]

OR

- (a) Describe the procedure of seismic refraction method for groundwater exploration. [5]
 Differentiate between : Unconfined aquifer and leaky aquifer [2]
 (b) Give an account of different types of aquifer with suitable sketches. [4]
 What is Water Harvesting? Enlist advantages of rooftop rainwater harvesting system. [3]
